

model provides an adequate fit that does not appear to improve with additional classes.” Here the PPN study also suggests the two-class mixture model is sufficient to model the data.

For infant i , denote their scores in each of the three tests as $\{\mathbf{x}_i^{(1)}, \mathbf{x}_i^{(2)}, \mathbf{x}_i^{(3)}\}$ and their group indicator by z_i . Following [Stern et al. \(1995\)](#), we assume that infants in group k will have the same score probabilities, $(\theta_k^{(1)}, \theta_k^{(2)}, \theta_k^{(3)})$, across the three tests. The multinomial mixture model with K groups is:

$$\pi \sim \text{Dirichlet}(\alpha_\pi \mathbf{1}_K), \quad (21)$$

$$\theta_k^{(j)} \sim \text{Dirichlet}(\alpha), \quad k = 1, \dots, K; \quad j = 1, 2, 3. \quad (22)$$

$$z_i \sim \text{Categorical}(\pi), \quad i = 1, \dots, n \quad (23)$$

$$\mathbf{x}_i^{(j)} | z_i, \theta \sim \text{Multinomial}(\theta_{z_i}^{(j)}). \quad (24)$$

We set $\alpha = 2$ and $\alpha_\pi = 2$, following the recommendation of [Gelman et al. \(1996\)](#) to use a “weak but not uniform prior distribution.” To draw from the posterior predictive, we use Gibbs sampling.

For both partial predictive checks and PPNs, we use the heldout diagnostic [Eq. 12](#). The underlying diagnostic function is the χ^2 -discrepancy:

$$d_K(\mathbf{x}, \theta) = 2 \sum_{i=1}^n \sum_{k=1}^K \sum_{j=1}^3 x_i^{(j)} \log \left(\frac{x_i^{(j)}}{\mathbb{E}[x_i^{(j)} | \theta]} \right), \quad (25)$$

$$\text{where } \mathbb{E}[x_i^{(j)} | \theta] = \sum_{k=1}^K \theta_k^{(j)} p(z_i = k | \mathbf{x}_i, \theta). \quad (26)$$

We split the data into $\mathbf{x} = \{\mathbf{x}_{\text{in}}, \mathbf{x}_{\text{out}}, \mathbf{x}_{\text{val}}\}$, where $\mathbf{x}_{\text{in}}$ is used to draw posterior predictive data, $\mathbf{x}_{\text{out}}$ is used as the out of sample data in the partial predictive check, and $\mathbf{x}_{\text{val}}$ is used to define the diagnostic. Specifically, the heldout diagnostic is:

$$d_K(\mathbf{x}; \mathbf{x}_{\text{val}}) = \mathbb{E}[d_K(\mathbf{x}, \theta) | \mathbf{x}_{\text{val}}], \quad (27)$$

which is approximated via Monte Carlo with samples from the posterior $p(\theta | \mathbf{x}_{\text{val}})$.

We consider mixture models with $K \in \{1, 2, 3, 4\}$ components. All four models pass their partial predictive checks ([Figure 5](#)); that is, all models generate predictive distributions which are consistent with the observed data (according to the heldout diagnostic). As we cannot eliminate models based on goodness-of-fit, we use a PPN study to determine which models are providing essentially the same predictions.

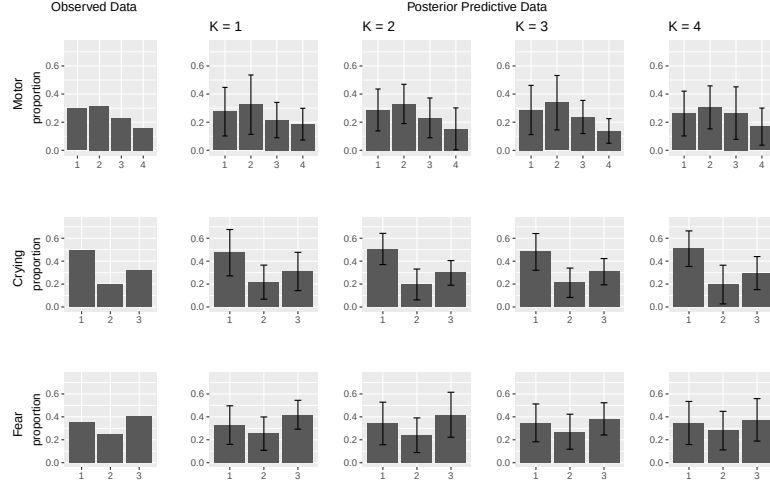


Figure 4: Posterior predictive draws from multinomial mixture models. A PPN study will help determine which models provide no improvement over other models. On the left are the observed data proportions. Beside it are datasets draws from the posterior predictive distributions of different mixtures $p_K(\mathbf{x}_{\text{rep}}|\mathbf{x}_{\text{in}})$ for $K \in \{1, 2, 3, 4\}$. Error bars are the 95% prediction intervals.

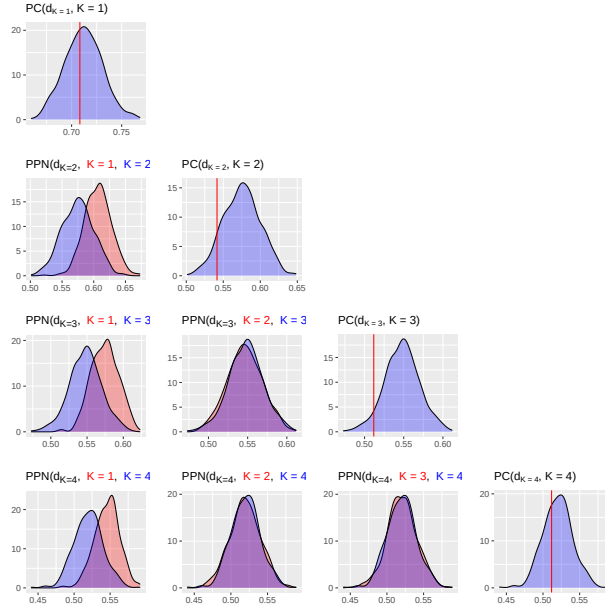


Table 1:: Symmetrized Kullback-Leibler distance between distributions.

	$K = 1$	$K = 2$	$K = 3$
$K = 1$			
$K = 2$	2.18		
$K = 3$	2.13	0.16	
$K = 4$	1.79	0.24	0.24

Figure 5: A PPN study of multinomial mixture models. This study suggests that $K = 2$ is consistent with the data (no further components are needed). On the diagonal are heldout predictive checks displaying $p_A(d_A(\mathbf{x}_{\text{rep}}^A; \mathbf{x}_{\text{val}})|\mathbf{x}_{\text{in}})$ (blue histogram) and $d_A(\mathbf{x}_{\text{out}}; \mathbf{x}_{\text{val}})$ (red line). All models pass the checks. To the left of the diagonal are PPNs, each one checking if a simpler model can fool the model under study. Specifically, the PPNs compare $p_B(d_A(\mathbf{x}_{\text{rep}}^B; \mathbf{x}_{\text{val}})|\mathbf{x}_{\text{in}})$ (red histogram) and $p_A(d_A(\mathbf{x}_{\text{rep}}^A; \mathbf{x}_{\text{val}})|\mathbf{x}_{\text{in}})$ (blue histogram). The PPN study shows that $K = 1$ passes its check but it does not fool $K = 2$. $K = 2$ passes its check and can fool $K = 3$ and 4.

The PPN study proceeds as follows.

- Based on visual inspection and the symmetrized Kullback-Leibler distance, the PPN suggests that $K = 1$ cannot fool $K = 2$.
- The PPN suggests that $K = 2$ can fool $K = 3$.
- The PPN suggests that $K = 2$ can fool $K = 4$.

The PPN study suggests $K = 2$ is adequate for modeling the data; increasing the number of mixture components beyond $K = 2$ is not justified. This finding corroborates that of Gelman et al. (1996), who made the heuristic choice of $K = 2$. The posterior predictive draws from the different models have high variability Figure 4, preventing direct visual comparison of predictive distributions at the data level.

We additionally compute the Bayes factors to compare the models. We approximate the marginal likelihood by the harmonic mean of the likelihood values (for details, see Appendix B of the Supplementary Material). The Bayes factors provide inconclusive evidence (Table 5 of the Supplementary Material).

3.2 Linear Factor Analysis

When is a nonlinear model required for factor analysis, and when is a linear model adequate? To investigate the capacity of a PPN study to help answer this question, we consider two simulation settings. In one, the data is generated from a linear factor model; in the other, it is generated from a nonlinear factor model.

The observed data is $\mathbf{x}_i \in \mathbb{R}^G$, $i = 1, \dots, N$. We assume that $\mathbf{x}_i$ has some low dimensional representation $\mathbf{z}_i \in \mathbb{R}^K$ with

$$\mathbf{x}_i = f(\mathbf{z}_i) + \boldsymbol{\varepsilon}_i \quad (28)$$

for some function $f : \mathbb{R}^K \rightarrow \mathbb{R}^G$ and noise term $\boldsymbol{\varepsilon}_i \in \mathbb{R}^G$. We consider three different modeling strategies for estimating this mapping between the latent representation and the observed data: (i) probabilistic principal components analysis (PPCA, Tipping and Bishop, 1999); (ii) a deep generative model, fit with a variational autoencoder (Kingma and Welling, 2014); and (iii) a deep generative model with skip connections, fit with a skip-VAE (Dieng et al., 2019).

Models

Probabilistic PCA (Tipping and Bishop, 1999) assumes that f is a linear mapping from the low-dimensional latent representation to the observed data,

$$\mathbf{x}_i = \mathbf{W}\mathbf{z}_i + \boldsymbol{\varepsilon}_i, \quad \boldsymbol{\varepsilon}_i \sim N(\mathbf{0}, \sigma^2 \mathbf{I}).$$

The latent variables are assigned a normal prior, $\mathbf{z}_i \sim N(0, \mathbf{I})$. Tipping and Bishop (1999) estimated the linear mapping $\mathbf{W}$ and representations $\mathbf{z}_i$ using the EM algorithm.